NANDHINI KALIAPPAN

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Location: Maynooth, Co. Kildare, Ireland

PROFESSIONAL SUMMARY

Data Engineer with 9+ years of experience in designing, building and maintaining large-scale data warehouses, data lakes and Lakehouse platforms. Skilled in end-to-end ETL/ELT pipeline development, cloud data platform architecture and data modelling for enterprise level application using Azure Data Factory, Databricks, Apache Spark/PySpark, dbt and Apache Airflow with strong expertise in Python, SQL, data modelling and data quality. Proven ability to architect robust, scalable and adaptable data solutions long term impact on platform and product strategy. Experienced in consulting cross-functional software and teams, design decisions made with industries best practices and emerging technologies. Holds an MSc in Data Science and Analytics from Maynooth University. Seeking a full-time Data Engineering role in Ireland.

SKILLS

Programming Languages: Python, SQL, PL/SQL, T-SQL, R, PySpark

Big Data Processing & Frameworks: Apache Spark, PySpark, Databricks, Delta Lake

Databases & Data Warehousing: MS SQL Server, Snowflake, Oracle, Teradata, MongoDB

ETL/ELT & Orchestration: Azure Data Factory (ADF), SSIS, Alteryx, DBT, Airflow, DAG Orchestration, Control-M, Informatica PowerCenter, Informatica Cloud (IICS)

Cloud Platform: Microsoft Azure (Synapse Analytics, Data Lake, Databricks, Azure SQL, Azure Pipelines)

BI & Visualisation: Power BI, Tableau, OBIEE, GIS

Data Modelling & Architecture: Dimensional Modelling (Star Schema), Data Vault, OLAP Design, Data Lake / Lakehouse Architecture, Real-time Data Pipelines, Event-driven Systems

Data Quality, Governance & Analytics: Great Expectations, Data Lineage, GDPR Awareness, ML Pipelines, Data Science fundamentals

DevOps, CI/CD & Tools: GitHub Actions, Azure Pipelines (CI/CD), JIRA, ServiceNow

Version Control: GitHub, Azure Repos

Certifications: Microsoft Azure DP-203 (Data Engineering Expert), Alteryx Designer Core, Microsoft Azure AZ-900, Microsoft Azure DP-900

WORK HISTORY

Assistant Manager | **Ernst & Young (EY)** – Chennai, India | September 2021 – August 2025

- Architected and delivered large-scale, multi-terabyte data warehouse and data lake solutions on Microsoft Azure and snowflake, supporting multiple business domains with designs built to scale with evolving business requirements.
- Designed and implemented an automated data backup and archival pipeline using Azure Databricks, optimising storage efficiency and reducing database infrastructure costs by 20%.
- Led end-to-end development of ETL/ELT pipelines using Azure Data Factory and Databricks (Apache Spark/PySpark) and snowflake, applying performance tuning techniques that cut processing time by 30%.
- Developed a modular dbt models on snowflake across staging, intermediate and mart layers, using incremental models, schema tests and auto-generated documentation to improve data quality, lineage and consistency for downstream analytics.
- Orchestrated dbt transformations and snowflake loads using Apache Airflow, designing DAG's with retries, scheduling and failure alerts to deliver reliable data workflows with minimal manual intervention.
- Served as an technical consultant to software engineering and data science teams — providing architectural guidance, design recommendations, and dimensional data modelling expertise.
- Researched and benchmarked emerging data technologies and industry solutions, informing architectural decisions and keep solutions aligned with industry best practices.
- Owned the full project lifecycle from requirements and design through testing, deployment, supporting UAT and Production issues through root-cause analysis, reducing incident resolution time.

- Led a team of 10 engineers, running scrum ceremonies, managing sprint priorities and resources, and mentoring members on technical best practice and professional growth.

Cognizant Technology Solutions – Coimbatore, India | October 2019 – September 2021

- Delivered enhancements, change requests and production support on ETL pipelines using Informatica PowerCenter, Teradata, OWB, OBIEE, Appworx, and Oracle DB.
- Developed Informatica mappings and workflows, performing unit testing and UAT testing to ensure quality deliverables.
- Supported ETL operations using Microsoft Azure SSIS, SSMS, and ServiceNow, including on a rotational shift basis.
- Performed production deployments and created automated reports for business stakeholders as part of process improvement initiatives.

Accenture Solutions – Chennai, India | May 2016 – October 2019

- Worked in fast-paced, lead driven development environment with a hands-on experience in the end-to-end project development lifecycle.
- Developed and maintained ETL pipelines using Informatica PowerCenter, Oracle, Salesforce, and Informatica Cloud (IICS) for Salesforce data integration.
- Implemented Control-M scheduling on the Siebel ETL application, automating job orchestration which reduced manual work by 50%.

EDUCATION

Master of Science in Data Science and Analytics (MSc)| Maynooth University, Ireland | 2025 – 2026

Bachelor of Engineering in Electrical and Electronics Engineering (BE) | Sri Krishna College of Engineering and Technology, India | 2013 – 2016

Diploma in Mechatronics Engineering | PSG College of Technology, India | 2010 – 2013